Proofs of Theorems

Jeremy Rosen

1 Portfolio Choice with Matriculation

Theorem 1. *For large N, the committee's portfolio choice problem with matriculation probabilities has a cutoff rule solution on success probabilities.*

Proof. For brevity, let $w = (x, z)$. Then for large N, the portfolio choice problem is:

$$\begin{aligned} &\max_{\delta \in \{\delta | \delta : \mathbb{R}^K \to [0,1]\}} \int_{w \in \mathbb{R}^K} P(y = S|w) P(m = M|w) \delta(w) f(w) dw, \\ &\text{where: } \int_{w \in \mathbb{R}^K} P(m = M|w) \delta(w) f(w) dw \leq N_M / N, \end{aligned} \tag{1}$$

where δ is the committee's decision rule, and f is the density of w. Because success and matriculation probabilities, as well as densities, are never negative, the unconstrained objective increases in δ, which means that the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned} \mathcal{L}(\delta; \lambda) &= \int_{w \in \mathbb{R}^K} P(y = S|w) P(m = M|w) \delta(w) f(w) dw \\ &\quad + \lambda \left\{ N_M / N - \left[\int_{w \in \mathbb{R}^K} P(m = M|w) f(w) dw - \int_{w \in \mathbb{R}^K} P(m = M|w)(1 - \delta(w)) f(w) dw \right] \right\} \\ &= \int_{w \in \mathbb{R}^K} \left[P(y = S|w) \delta(w) + \lambda (1 - \delta(w)) \right] P(m = M|w) f(w) dw \\ &\quad + \lambda \left[N_M / N - \int_{w \in \mathbb{R}^K} P(m = M|w) f(w) dw \right]. \end{aligned} \tag{2}$$

In the discrete case, there is not necessarily a cutoff rule solution. But in the continuous case, the committee can accept just enough applicants so that $\int_{w \in \mathbb{R}^K} P(m = M|w) \delta(w|\lambda^*) f(w) dw = N_M / N$, which yields a cutoff rule solution on success probabilities alone:

$$\forall w \in \mathbb{R}^K, \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y = S|w) > \lambda^* \\ [0,1] & \text{if } P(y = S|w) = \lambda^* \\ 0 & \text{if } P(y = S|w) < \lambda^*. \end{cases} \tag{3}$$

The committee accepts all applicants with success probabilities $P(y = S|w)$ strictly greater than the cutoff probability λ^*, rejects all applicants with $P(y = S|w)$ strictly less than λ^*, and may accept

or reject any applicants with $P(y = S|w)$ equal to λ^*, if such a mass of applicants exists. All that remains is to identify λ^*. Like Bai et al. (2022), I define the following two functions:

$$\begin{aligned} M^H(\lambda) &= \int_{\{w \in \mathbb{R}^K | P(y=S|w) \geq \lambda\}} P(m = M|w) f(w) dw \\ M^L(\lambda) &= \int_{\{w \in \mathbb{R}^K | P(y=S|w) > \lambda\}} P(m = M|w) f(w) dw. \end{aligned} \tag{4}$$

The upper bound function M^H is the fraction of applicants, scaled by their matriculation probabilities, whose success probabilities are greater than or equal to some λ, and the lower bound function M^L is the fraction whose success probabilities are strictly greater than λ. Both functions are between 0 and 1, both are decreasing in λ, and for any λ, it must be that $M^H(\lambda) \geq M^L(\lambda)$.

Then it is the case that $\lambda^* = \inf_{\lambda} \{\lambda | M^L(\lambda) \leq N_M/N\}$. Intuitively, λ^* is the smallest cutoff probability where the fraction of matriculation-probability-scaled applicants with $P(y = S|w) > \lambda$ does not exceed N_M/N. If there is a positive mass of matriculation-probability-scaled applicants with $P(y = S|w) = \lambda^*$ (that is, if $M^H(\lambda^*) - M^L(\lambda^*) > 0$), then the committee will randomly accept the exact fraction $\alpha^* \in [0,1]$ of that mass of applicants such that $M^L(\lambda^*) + \alpha^*[M^H(\lambda^*) - M^L(\lambda^*)] = N_M/N$. That is, the committee will accept every applicant with $P(y = S|w) > \lambda^*$ and accept just enough applicants with $P(y = S|w) = \lambda^*$ to fill the cohort. □

2 Cutoff Probability Identification

Theorem 2. *For simplicity, suppose there is one round. The maximum likelihood estimate $\hat{P}_{y^*} = \hat{P}(y^* = S|t)$ is such that the number of acceptances N_A equals the expected N_A. Also, for any $\hat{P}_{y^*}$ and N_A, $\mathcal{L}^R$ is highest when the committee accepts the N_A highest $P_{y_n} = P(y_n = S|x_n, z_n, t_n, \bar{k})$.*

Proof. Supposing one round, the restricted likelihood is:

$$\begin{aligned} \mathcal{L}^R = \prod_{n=1}^{N} &\left[\frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}}\right]^{\mathbb{1}(d_n=A)} \left[\frac{e^{P_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}}\right]^{\mathbb{1}(d_n=R)} \\ &\left[\frac{e^{P_{y_n}}}{1+e^{P_{y_n}}}\right]^{\mathbb{1}(y_n=S)} \left[\frac{1}{1+e^{P_{y_n}}}\right]^{\mathbb{1}(y_n=F)} \end{aligned} \tag{5}$$

When an applicant is accepted, the resulting term in $\mathcal{L}^R$ decreases in P_{y^*}, providing downward pressure on $\hat{P}_{y^*}$. Also, when one is rejected, the term increases in P_{y^*}, providing upward pressure.

Since $\mathbb{1}(d_n = A) + \mathbb{1}(d_n = R) = \mathbb{1}(y_n = S) + \mathbb{1}(y_n = F) = 1$, the restricted log-likelihood is:

$$\begin{aligned}
\ell^R &= \sum_{n=1}^{N} \left[\mathbb{1}(d_n = A) \log\left(\frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}} \right) + \mathbb{1}(d_n = R) \log\left(\frac{e^{P_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}} \right) \right. \\
&\quad \left. + \mathbb{1}(y_n = S) \log\left(\frac{e^{P_{y_n}}}{1 + e^{P_{y_n}}} \right) + \mathbb{1}(y_n = F) \log\left(\frac{1}{1 + e^{P_{y_n}}} \right) \right] \\
&= \sum_{n=1}^{N} \left[\frac{1}{\sigma} \left(\mathbb{1}(d_n = A) P_{y_n} + \mathbb{1}(d_n = R) P_{y^*} \right) - \log\left(e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma} \right) \right. \\
&\quad \left. + \mathbb{1}(y_n = S) P_{y_n} - \log\left(1 + e^{P_{y_n}} \right) \right].
\end{aligned} \tag{6}$$

Since ℓ^R is strictly concave in P_{y^*}, I find the unique maximizer $\hat{P}_{y^*}$ by the first-order condition:

$$\frac{1}{\sigma} \left[N - N_A - \sum_{n=1}^{N} \frac{e^{\hat{P}_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{\hat{P}_{y^*}/\sigma}} \right] = 0 \quad \Rightarrow \quad \sum_{n=1}^{N} \frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{\hat{P}_{y^*}/\sigma}} = N_A. \tag{7}$$

The left side of the latter equation is each applicant's acceptance probability summed over the number of applicants; that is, expected N_A. Therefore, $\hat{P}_{y^*}$ is such that the expected N_A equals N_A.

Given that $\hat{P}_{y^*}$ is identified, I will show that ℓ^R, and therefore $\mathcal{L}^R$, is highest when the committee accepts the N_A applicants with the highest P_{y_n}. The only part of ℓ^R that depends on d_n is $\sum_{n=1}^{N} [\mathbb{1}(d_n = A) P_{y_n} + \mathbb{1}(d_n = R) P_{y^*}] = N P_{y^*} + \sum_{n=1}^{N} \mathbb{1}(d_n = A)(P_{y_n} - P_{y^*})$; this equality holds since $\mathbb{1}(d_n = A) + \mathbb{1}(d_n = R) = 1$. As such, the only part depending on d_n is $\sum_{n=1}^{N} \mathbb{1}(d_n = A)(P_{y_n} - P_{y^*})$.

This expression is highest when $d_n = A$ for the N_A highest values of P_{y_n}, so the likelihood is highest when the committee accepts the N_A highest P_{y_n}. Intuitively, the committee's selectivity identifies $\hat{P}_{y^*}$, and it is not necessary for identification for the committee to accept applicants with higher P_{y_n}. But if the committee engages in positive sorting, the model's fit will improve. □

3 Deviation Gains Test p-Value

Theorem 3. *For two sets that are not chosen using $\hat{\theta}_S^U$, as $B \to \infty$, the fraction of perturbations in which the model's set does no better than the committee's, $g(\tilde{\theta}_{S,b}^U) \leq 0$, converges to the p-value of the test of $H_0 \colon g_0 = 0$ vs. $H_1 \colon g_0 > 0$ that uses the observed gain $\hat{g}$ as the test statistic.*

Proof. Let g be the deviation gain defined in the paper, with both the model's set and the committee's set held fixed. Since the two sets are determined before performing the test, the proof works the same way for matriculation sets as for acceptance sets. Suppose the maximum likelihood estimate $\hat{\theta}_S^U \overset{a}{\sim} \mathcal{N}(\theta_S^0, \Sigma_0)$, and each perturbation draw $\tilde{\theta}_{S,b}^U \sim \mathcal{N}(\hat{\theta}_S^U, \hat{\Sigma})$, where $\hat{\Sigma}$ is consistent for Σ_0. Because g is a smooth function of θ_S, it is true by the delta method that $\hat{g} = g(\hat{\theta}_S^U) \overset{a}{\sim} \mathcal{N}(g_0, \sigma_g^2)$ and $\tilde{g} = g(\tilde{\theta}_{S,b}^U) \mid \hat{g} \overset{a}{\sim} \mathcal{N}(\hat{g}, \sigma_g^2)$, where $g_0 = g(\theta_S^0)$ and $\sigma_g^2 = \nabla g(\theta_S^0)' \Sigma_0 \nabla g(\theta_S^0)$.

The test statistic is the observed gain $\hat{g}$, and its p-value is the probability, if the null is true, of an estimated gain at least as large as the observed gain. Denote $\hat{g}' = \sigma_g Z$ as a hypothetical estimated

gain generated under $g_0 = 0$, where $Z \sim \mathcal{N}(0,1)$. Then, using $P(Z \geq c) = \Phi(-c)$:

$$P(\hat{g}' \geq \hat{g} \mid g_0 = 0) = P(Z \geq \hat{g}/\sigma_g) = \Phi(-\hat{g}/\sigma_g) = 1 - \Phi(\hat{g}/\sigma_g). \tag{8}$$

As the number of perturbations B becomes large, it is true by the law of large numbers that the fraction of perturbations with $\tilde{g} \leq 0$ converges to the probability that a single perturbation is weakly negative. As such, for large B, this fraction, which I denote as $\hat{p}$, is the p-value:

$$\hat{p} = P(\tilde{g} \leq 0 \mid \hat{g}) = \Phi\left(\frac{0 - \hat{g}}{\sigma_g}\right) = \Phi(-\hat{g}/\sigma_g) = 1 - \Phi(\hat{g}/\sigma_g). \tag{9}$$

Finally, although the null hypothesis is $g_0 = 0$, the test is also valid, and conservative, if $g_0 < 0$, which can occur because a fixed model set need not be better than the committee's set at the true parameters. That is, $P(\hat{p} \leq \alpha) \leq \alpha$ whenever $g_0 \leq 0$. Because inverses of strictly increasing functions preserve inequalities, Φ is strictly increasing, and $\hat{p} = \Phi(-\hat{g}/\sigma_g)$, it is the case that $\hat{p} \leq \alpha$ holds exactly when $\hat{g} \geq -\sigma_g \Phi^{-1}(\alpha)$. Substituting $\hat{g} = g_0 + \sigma_g Z$:

$$P(\hat{p} \leq \alpha) = P(Z \geq -\Phi^{-1}(\alpha) - g_0/\sigma_g) = \Phi(\Phi^{-1}(\alpha) + g_0/\sigma_g). \tag{10}$$

For $g_0 = 0$, it is the case that $P(\hat{p} \leq \alpha) = \Phi(\Phi^{-1}(\alpha)) = \alpha$. For $g_0 < 0$, because Φ is strictly increasing, it is the case that $P(\hat{p} \leq \alpha) = \Phi(\Phi^{-1}(\alpha) + g_0/\sigma_g) < \alpha$. Therefore, rejecting H_0 when $\hat{p} \leq \alpha$ produces Type 1 errors, i.e. false rejections, with probability at most α. □

4 Parametric Bootstrap Correction

Theorem 4. Under a linear probability model, it is the case that:

$$\begin{aligned} \mathbb{E}\big[\Delta(\hat{\theta}_S^U; \theta_S^0)\big] &= \mathbb{E}\big[\overline{P}^{\max}(\beta_0 + \eta)\big] - \overline{P}^{\max}(\beta_0) \qquad \text{and} \\ \mathbb{E}\big[\overline{\Delta}_B\big] &= \mathbb{E}\big[\overline{P}^{\max}(\beta_0 + \sqrt{2}\eta)\big] - \mathbb{E}\big[\overline{P}^{\max}(\beta_0 + \eta)\big], \end{aligned} \tag{11}$$

where the first expectation is over the sampling error and the second over the sampling error and perturbation draws. The winner's curse and its estimate are successive increments of $\mathbb{E}[\overline{P}^{\max}(\beta_0 + \sqrt{\tau}\eta)]$, from $\tau = 0$ to 1 and from 1 to 2. The correction is exact when that function is linear in τ on $[0,2]$, too small when it is concave in τ there, and too large when it is convex in τ there.

Proof. For the first identity, at the estimates, the model's set attains $\overline{P}^{\max}(\hat{\beta}^U) = \overline{P}^{\max}(\beta_0 + \eta)$. The optimal set $\mathcal{A}^R(\theta_S^0)$, which is fixed, attains $\overline{w}'_{\mathcal{A}^R(\theta_S^0)}(\beta_0 + \eta)$, with expectation $\overline{P}^{\max}(\beta_0)$. For the second identity, the set chosen at the perturbation and evaluated there attains $\overline{P}^{\max}(\tilde{\beta}^U) = \overline{P}^{\max}(\beta_0 + \eta + (\tilde{\beta}^U - \hat{\beta}^U))$. η and $\tilde{\beta}^U - \hat{\beta}^U$ are independent, each normal with mean zero and the same covariance, so their sum is normal with mean zero and twice that covariance, which is also the distribution of $\sqrt{2}\eta$. Therefore, $\mathbb{E}[\overline{P}^{\max}(\tilde{\beta}^U)] = \mathbb{E}[\overline{P}^{\max}(\beta_0 + \sqrt{2}\eta)]$. The set chosen at $\hat{\theta}_S^U$ is a function of η alone, so at the perturbation, it attains $\overline{w}'_{\mathcal{A}^R(\hat{\theta}_S^U)}\tilde{\beta}^U$. This quantity is $\overline{P}^{\max}(\beta_0 + \eta)$ plus $\overline{w}'_{\mathcal{A}^R(\hat{\theta}_S^U)}(\tilde{\beta}^U - \hat{\beta}^U)$, a term with expectation zero given η. Intuitively, the winner's curse is

how much the best-looking average success probability improves when one η of noise is added to the truth, and the estimated correction is how much more it improves when a second, independent η of noise is added on top. The correction is exact when the second increment equals the first, too small when it is smaller (i.e. concave in τ), and too large when it is larger (i.e. convex in τ). □

Corollary 1. If $\beta_0 = 0$, so that every applicant has the same true success probability, then:

$$\mathbb{E}\big[\hat{g}\big] = \mathbb{E}\big[\overline{P}^{\max}(\eta)\big] \qquad \text{and} \qquad \mathbb{E}\big[\Delta(\tilde{\theta}_S^U; \hat{\theta}_S^U)\big] = (\sqrt{2}-1)\,\mathbb{E}\big[\overline{P}^{\max}(\eta)\big]. \tag{12}$$

Here, the estimated correction has expectation $(\sqrt{2}-1) \approx 0.414$ times the winner's curse. This is the worst-case scenario, meaning that in every data-generating process,

$$\mathbb{E}[\overline{\Delta}_B] \geq (\sqrt{2}-1)\,\mathbb{E}[\Delta(\hat{\theta}_S^U; \theta_S^0)].$$

Proof. With $\beta_0 = 0$, it is the case that $\overline{P}^{\max}(\beta_0 + \sqrt{\tau}\eta) = \sqrt{\tau}\overline{P}^{\max}(\eta)$ because $\overline{P}^{\max}$ is homogeneous of degree one. Therefore, the two increments of Theorem 4 are $\mathbb{E}[\overline{P}^{\max}(\eta)]$ and $(\sqrt{2}-1)\mathbb{E}[\overline{P}^{\max}(\eta)]$. In addition, the committee's set is fixed, so the observed gain $\hat{g} = \overline{P}^{\max}(\eta) - \overline{w}'_{\mathcal{A}^U}\eta$ has expectation $\mathbb{E}[\overline{P}^{\max}(\eta)]$, which equals the winner's curse when $\beta_0 = 0$. As such, the estimated correction in this scenario has expectation $(\sqrt{2}-1) \approx 0.414$ times the winner's curse.

For the claim that this is the worst-case scenario, fix η and view $\overline{P}^{\max}(\beta_0 + \sqrt{\tau}\eta) = \max_{\mathcal{A}}[\overline{w}'_{\mathcal{A}}\beta_0 + \sqrt{\tau}\overline{w}'_{\mathcal{A}}\eta]$ as a function of $\sqrt{\tau}$. Each term inside the maximum is linear in $\sqrt{\tau}$, a maximum of linear functions is convex, and averaging over η preserves convexity. Therefore, $\mathbb{E}[\overline{P}^{\max}(\beta_0 + \sqrt{\tau}\eta)]$ is convex in $\sqrt{\tau}$. A convex function lies below the chord through any two of its points. Also, the values of $\mathbb{E}[\overline{P}^{\max}(\beta_0 + \sqrt{\tau}\eta)]$ at $\sqrt{\tau} = \{0, 1, \sqrt{2}\}$ are $\overline{P}^{\max}(\beta_0)$, $\mathbb{E}[\overline{P}^{\max}(\beta_0 + \eta)]$, and $\mathbb{E}[\overline{P}^{\max}(\beta_0 + \sqrt{2}\eta)]$. Finally, $1 = (1 - 1/\sqrt{2}) \cdot 0 + (1/\sqrt{2}) \cdot \sqrt{2}$ lies between 0 and $\sqrt{2}$ with weights $1 - 1/\sqrt{2}$ and $1/\sqrt{2}$, which yields the inequality:

$$\mathbb{E}\big[\overline{P}^{\max}(\beta_0 + \eta)\big] \leq (1 - 1/\sqrt{2})\,\overline{P}^{\max}(\beta_0) + (1/\sqrt{2})\,\mathbb{E}\big[\overline{P}^{\max}(\beta_0 + \sqrt{2}\eta)\big]. \tag{13}$$

Multiplying by $\sqrt{2}$ and rearranging:

$$\mathbb{E}\big[\overline{P}^{\max}(\beta_0 + \sqrt{2}\eta)\big] - \mathbb{E}\big[\overline{P}^{\max}(\beta_0 + \eta)\big] \geq (\sqrt{2}-1)\Big[\mathbb{E}\big[\overline{P}^{\max}(\beta_0 + \eta)\big] - \overline{P}^{\max}(\beta_0)\Big], \tag{14}$$

the two increments of Theorem 4. At $\beta_0 = 0$, this holds with equality, so it is the worst case. □

5 Portfolio Choice with Shocks

Theorem 5. *For large N, the committee's portfolio choice problem with preference shocks has a cutoff rule solution with logistic conditional choice probabilities.*

Proof. The portfolio choice problem with type 1 extreme value preference shocks is:

$$\begin{gathered}\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \big[(P(y_n = S|x_n, z_n) + \varepsilon_n(A))\mathbb{1}(d_n = A) + \varepsilon_n(R)\mathbb{1}(d_n = R)\big], \\ \text{where } \sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A.\end{gathered} \tag{15}$$

Because success probabilities are never negative, the unconstrained objective increases in expectation every time $d_n = A$, and so the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned}\mathcal{L}(d_1,\ldots,d_N;\lambda) &= \sum_{n=1}^{N}\left[(P(y_n = S|x_n,z_n)+\varepsilon_n(A))\mathbb{1}(d_n = A)+\varepsilon_n(R)\mathbb{1}(d_n = R)\right] \\ &\quad +\lambda\left\{N_A - \left[N - \sum_{n=1}^{N}\mathbb{1}(d_n = R)\right]\right\} \\ &= \sum_{n=1}^{N}\left[(P(y_n = S|x_n,z_n)+\varepsilon_n(A))\mathbb{1}(d_n = A)+(\lambda+\varepsilon_n(R))\mathbb{1}(d_n = R)\right] \\ &\quad +\lambda(N_A - N).\end{aligned} \tag{16}$$

Suppose without loss of generality that the applicants are sorted in decreasing order of their shock-perturbed success probabilities such that $P(y_1 = S|x_1,z_1)+\varepsilon_1(A)-\varepsilon_1(R) \geq \cdots \geq P(y_N = S|x_N,z_N)+\varepsilon_N(A)-\varepsilon_N(R)$. To ensure that the committee accepts exactly N_A applicants, it must be the case that $\lambda^* = \lambda^*(\boldsymbol{\varepsilon}(A),\boldsymbol{\varepsilon}(R)) \in \left[P(y_{N_A+1} = S|x_{N_A+1},z_{N_A+1})+\varepsilon_{N_A+1}(A)-\varepsilon_{N_A+1}(R), P(y_{N_A} = S|x_{N_A},z_{N_A})+\varepsilon_{N_A}(A)-\varepsilon_{N_A}(R)\right]$. That is, in this setup, λ^* is set identified. Then the solution is to accept all applicants whose shock-perturbed success probability exceeds λ^* and reject the rest:

$$\forall n \in \{1,\ldots,N\},\ d_n(x_n,z_n,\varepsilon_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n,z_n)+\varepsilon_n(A) > \lambda^*+\varepsilon_n(R) \\ \{A,R\} & \text{if } P(y_n = S|x_n,z_n)+\varepsilon_n(A) = \lambda^*+\varepsilon_n(R) \\ R & \text{if } P(y_n = S|x_n,z_n)+\varepsilon_n(A) < \lambda^*+\varepsilon_n(R). \end{cases} \tag{17}$$

This cutoff rule holds conditional on the realized shocks. Given the shocks, the committee computes λ^* and applies the cutoff, so there is no difficulty for the committee. But there is a complication regarding estimation, namely that unlike the committee, the econometrician does not observe the shocks. Therefore, the conditional choice probabilities used in estimation require taking expectations across each applicant's shocks, which would be fine except λ^* is a function of each applicant's shocks simultaneously. Suppose for now that λ^* is instead deterministic. Then:

$$\begin{aligned}P(d_n = A|x_n,z_n) &= P(P(y_n = S|x_n,z_n)+\varepsilon_n(A) \geq \lambda^*+\varepsilon_n(R)) \\ &= P(\varepsilon_n(R)-\varepsilon_n(A) \leq P(y_n = S|x_n,z_n)-\lambda^*) \\ &= \frac{\exp(P(y_n = S|x_n,z_n)/\sigma)}{\exp(P(y_n = S|x_n,z_n)/\sigma)+\exp(\lambda^*/\sigma)},\end{aligned} \tag{18}$$

and the likelihood becomes the product of individual binary logistics. The third equality holds because the difference of two independent T1EV random variables has a logistic distribution.

That said, λ^* is not deterministic for finite N, so the logistic conditional choice probabilities do not hold exactly. But for large N, λ^* converges to a constant. Denote $\hat{F}_N(c) = \frac{1}{N}\sum_{n=1}^{N}\mathbb{1}\{P(y_n = S|x_n,z_n)+\varepsilon_n(A)-\varepsilon_n(R) \leq c\}$ as the empirical distribution of the shock-perturbed success probabilities. The cutoff λ^* is the $(1-N_A/N)$ quantile of this distribution. Because the shocks are i.i.d., the perturbed indices in $\hat{F}_N$ are i.i.d. So by the Glivenko-Cantelli theorem, $\hat{F}_N$ converges almost surely to the population distribution F for large N. Therefore, as $N_A/N \to q$, $\hat{F}_N$'s $(1-N_A/N)$

quantile converges to F's deterministic $(1-q)$ quantile. The i.i.d. assumption is essential, and not only because each applicant's shock individually moves the cutoff by $O(1/N)$. A common perturbation component across applicants would have the same vanishing per-applicant influence, but it would slide the entire empirical distribution in unison, leaving λ^* random in the limit. Independence makes the simultaneous fluctuations of the N shocks cancel, so $\hat{F}_N$ and λ^* converge. With λ^* deterministic, no individual applicant's shock affects λ^*, the expectation defining each conditional choice probability factors across applicants, and the above logistic form holds. □